Visual Document Understanding

Hierarchical Table Extraction & Business Insights

Advanced Computer Vision for Deep Learning — Final Project

Team Members:

Lawrence Lin

Ryan Chen

Kacey Zhu

Jared Maksoud

Presentation Outline

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FinTabNet: 32,670 financial tables analyzed

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Problem Statement

The Challenge

Financial documents contain tables with:

- Multi-level headers & merged cells
- Irregular layouts (wide, tall, borderless)
- Low contrast & poor scan quality

Traditional OCR extracts text but loses the logical table structure, making downstream analytics impossible.

End-to-End Pipeline

1 Document Image

2 Table Structure Recognition (TSR) → HTML Structure

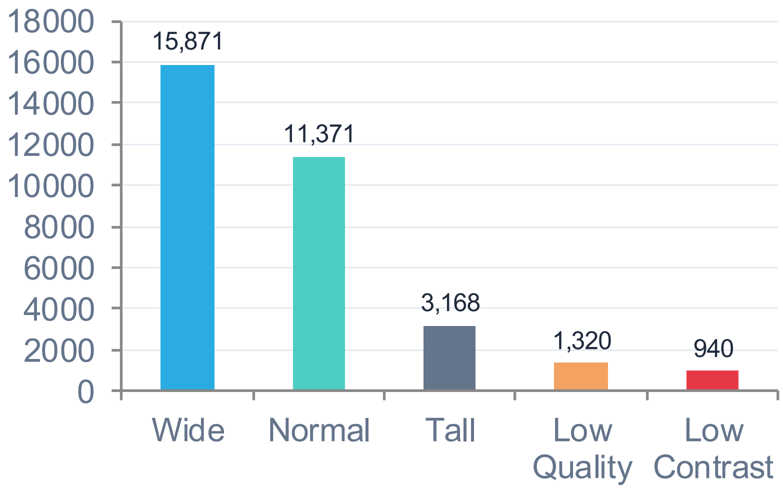
3 HTML → DataFrame

4 Key Information Extraction (KIE) → Business Records

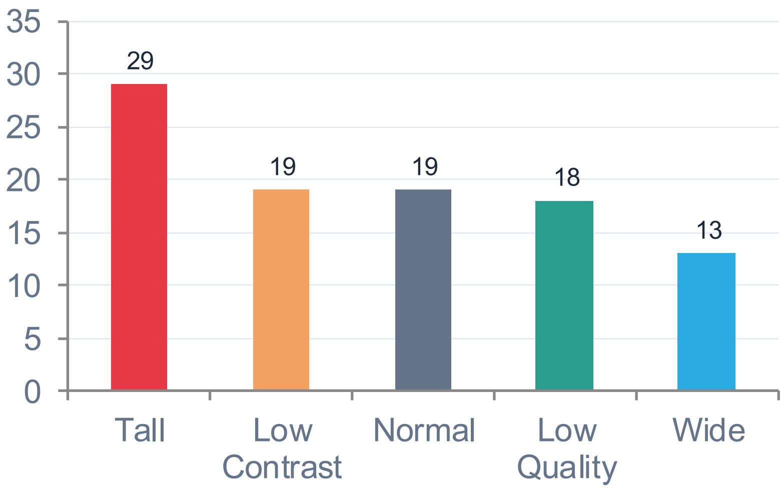
Dataset: FinTabNet EDA — Overview & Distribution

32,670	**850**	**150**	**300**
Total Records	Train split	Validation split	Test split

Image Type Distribution



Avg Hard Score by Image Type



Key finding: Wide tables dominate the dataset (48.6%), but tall tables score hardest (avg difficulty 29). Image type — not image quality — is the primary difficulty axis, driving our stratified sampling strategy.

Dataset: FinTabNet EDA — Structural Complexity

Structural Feature Ranges

Feature	Range	Mean / Median	Implication
num_rows	2 – 83	Mean 15.1	Long sequences → truncation risk
num_cells	4 – 963	Mean 75.1	Dense tables amplify alignment errors
max_colspan	1 – 25	**Median 3**	**Grouped headers universal**
max_rowspan	1 – 9	Median 1	Rare but disrupts row hierarchy
aspect_ratio	0.38 – 21.4	Median 2.93	**Wide layout = compressed columns**

Span Distribution

colspan > 1

98.1%

of all tables

Most = 2 or 3

rowspan > 1

6.5%

of all tables

Critical for row hierarchy

Key Correlations

r = 0.91

Taller image = more rows = harder

r = 0.87

Row count drives difficulty score

r = 0.85

Dense tables more error-prone

r ≈ 0

Image quality barely matters

r ≈ 0

Low contrast not the bottleneck

Key finding: Structural complexity — merged headers (98.1% colspan), row count, and cell density — is the core challenge. Image quality is largely irrelevant. Structure-aware metrics (TEDS) and hard examples are both essential for fair evaluation.

Dataset & Evaluation

Sampling

Stratified sampling from hard examples · **260 images × 5 table types** (wide, tall, low-contrast, normal, low-quality)

Preprocessing

- Standardize images to consistent resolution & format
- Clean HTML annotations — remove redundant tags
- Filter corrupted or incomplete samples
- Validate image–HTML pair alignment & consistency

Augmentation

Training only

Small-angle rotations: simulate scanning misalignment
Scaling: handle resolution variability
Contrast adjustment: model lighting differences
Blur simulation: mimic low-quality / noisy document images

Data Splits

850	150	300
Train	Val	Test

TEDS — Tree Edit Distance Similarity

$$\mathrm{TEDS} = 1 - \frac{\mathrm{TED}(T_{\mathrm{pred}}, T_{\mathrm{true}})}{\max(|T_{\mathrm{pred}}|, |T_{\mathrm{true}}|)}$$

Score 0–1 where 1.0 = perfect match. Structural errors (tag, colspan, rowspan) cost **1.0**; content mismatches cost **0.5**. Normalized by tree size — penalizes errors proportionally. Evaluated in two modes: Skeleton TEDS (structure only) and Pipeline TEDS (full output).

KIE — Key Information Extraction

Evaluated via entity-level F1 — measures correctness and completeness of extracted structured information.

Model Architectures Overview

UniTable

[Trainable]

Pixel-to-text encoder-decoder with ConvStem + Transformer. 3-pass pipeline: structure → bbox → content. LoRA fine-tuned.

Florence-2

[Trainable]

Unified vision-language model with prompt-based generation. LoRA fine-tuned for TSR and KIE subtasks.

SPARTAN

[Heuristic CV]

No trainable weights. Classical CV: adaptive thresholding → OCR clustering → grid reconstruction. CPU-only.

LlamaParse

[External API]

Production document parsing API integrating OCR + layout analysis. Industrial-grade, black-box.

ChatGPT

[External API]

GPT-4o-mini multimodal API. Prompt-based instruction reasoning. Strong semantic adaptability.

Model Performance Results

Model	Skeleton TEDS	Pipeline TEDS	F1 Score
UniTable (Fine-tuned)	0.8353 ⭐	0.3417	0.4928
UniTable (Original)	0.8353	0.3389	0.4829
SPARTAN	0.7253	0.5391 ⭐	0.70 ⭐
Florence-2 (LoRA)	0.0441	0.0287	N/A
ChatGPT	N/A	0.5109	N/A
LlamaParse	N/A	0.5120	N/A

⭐ = *Best in category among all models*

Florence-2: Zero-shot Strength vs Fine-tuning Challenges

Fine-tuning Setup

AdamW	1e-4	LoRA
Optimizer	LR	Method

Epochs: **50** — early stopped at epoch **17** *(unstable loss)*

Subtask Decomposition

Direct HTML generation proved too challenging for Florence-2. The task was decomposed into two subtasks:

TSR	KIE
HTML Skeleton	Cell Content

TEDS Results

Fine-tuned (LoRA)

0.0441	0.0287
Skeleton TEDS	Pipeline TEDS

Zero-shot baseline — Pipeline TEDS: **0.1539** ▲ *outperforms fine-tuned*

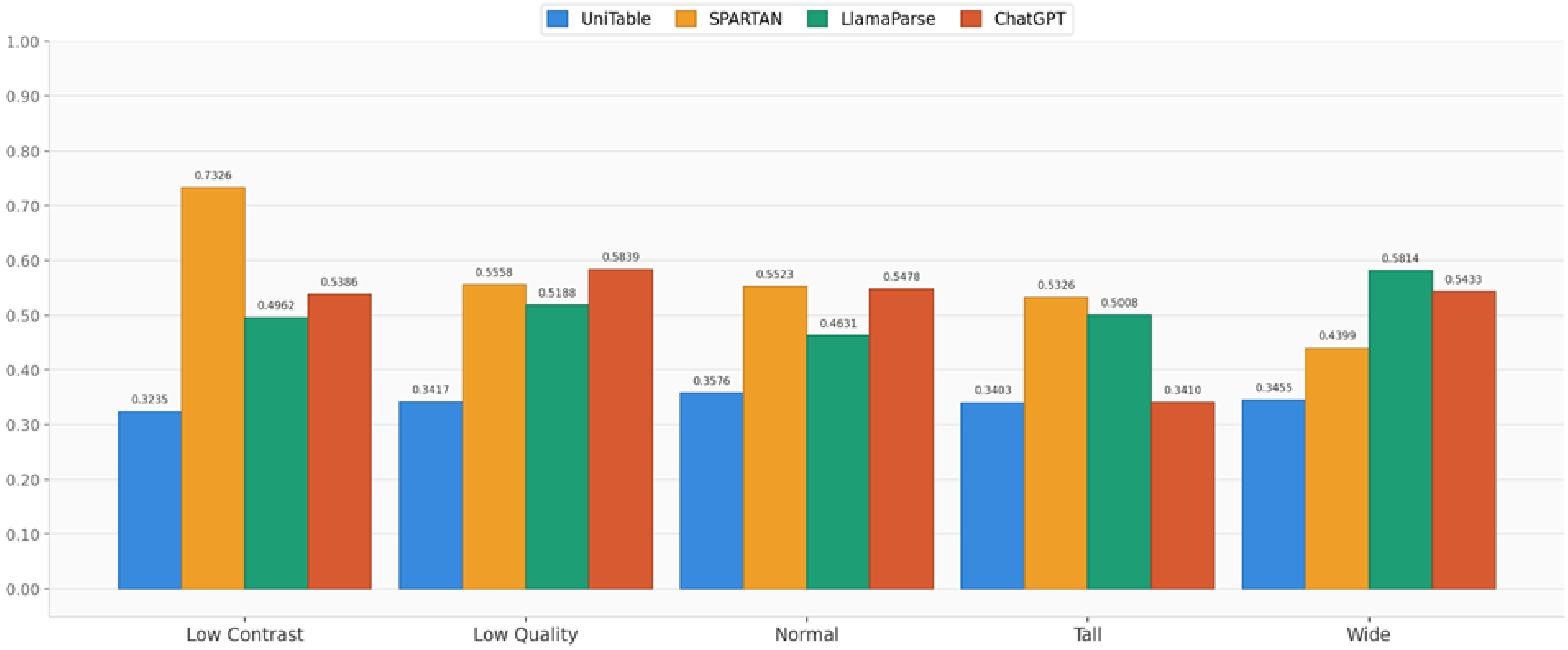
Why Fine-tuned Underperforms Zero-shot

Task Mismatch
Florence-2 pre-trained for OCR, captioning & grounding — not HTML generation. Large domain gap makes the task inherently difficult to learn.

Training Instability
Cross-entropy loss did not decrease steadily — model likely failed to converge, triggering early stopping at epoch 17 of 50.

LoRA Misconfiguration
Suboptimal rank/alpha may have given the adapter insufficient capacity or destabilized training through excessive parameter updates.

Pipeline TEDS by Image Type



SPARTAN leads on Low Contrast | ChatGPT collapses on Tall tables | LlamaParse most consistent

UniTable: Structure and Fine Tune

Fine-tuning Setup

AdamW
Optimizer

5e-5
LR

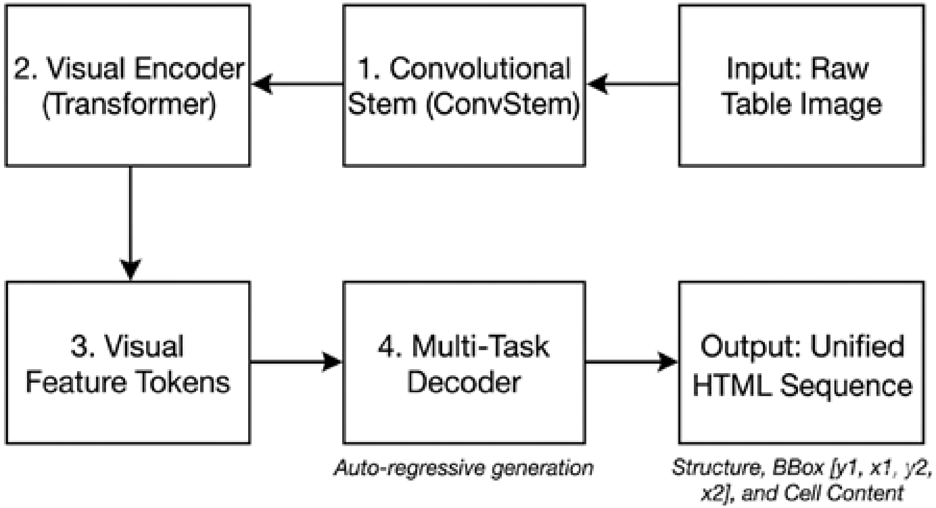
LoRA
Method

3
Epoch

Pass 1 — **Structure Model**
HTML skeleton with [] placeholders

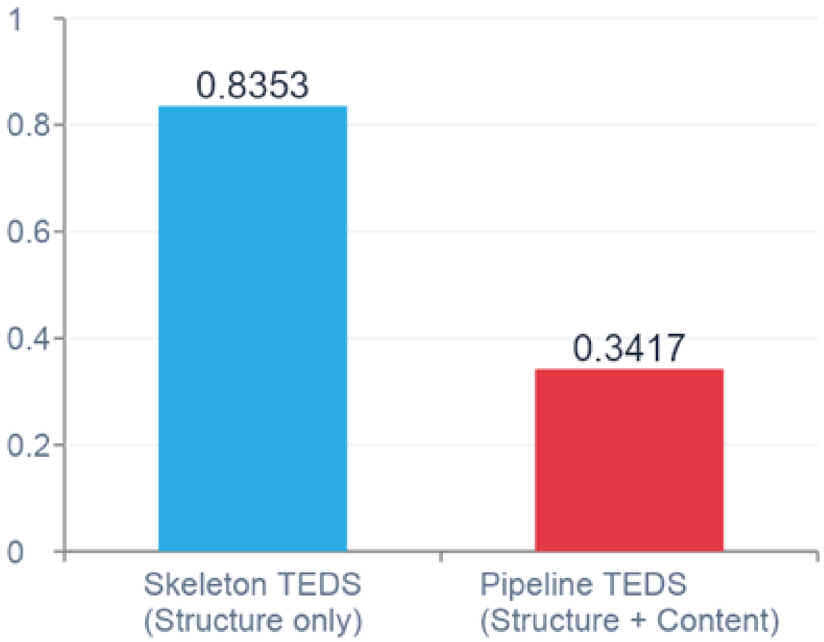
Pass 2 — **BBox Model**
Cell coordinates [y1, x1, y2, x2]

Pass 3 — **Content Model**
OCR fills placeholders



Vector Quantized - Variational Autoencoder (VQVAE): Original Embeddings compared to codebook to make the quantized loss minimum.

UniTable: Result



Root Cause:

Cascading errors — a single mispredicted structural token shifts every downstream cell.

SPARTAN: The Heuristic Surprise

Skeleton TEDS

UniTable
0.8353

SPARTAN
0.7253

Pipeline TEDS

UniTable
0.3417

SPARTAN
0.5391

Cell F1

UniTable
0.4928

SPARTAN
0.7

Why SPARTAN wins on Pipeline TEDS

No cascading errors: content is geometrically tied to structure — a wrong row shifts only that word, not the whole table

CPU-only, deterministic: runs in seconds, errors are traceable to OCR or geometry

Best on low contrast: Pipeline TEDS 0.7326 — highest of any model in that category

Weakness — span loss: 54.2% of SPARTAN failures involve lost column spans (multi-level headers)

+0.22 mean Pipeline TEDS gap over UniTable across all 5 image types

Error Analysis: Failure Modes

UniTable

- Structural token misprediction cascades to all subsequent cells
- Hallucinated content tokens corrupt full-row sequence
- Sequence length overflow on large tables (truncated decode)

Florence-2

- Severe structural oversimplification (predicted skeleton ≈ 2 cells)
- Incorrect text placed in wrong cells penalized by TEDS
- General VLM not pre-trained for markup generation

SPARTAN

- Span loss: merges grouped header columns (54.2% of failures)
- 80% of FinTabNet tables are borderless — LSD grid fails
- No learning: can't generalize beyond fixed heuristics

ChatGPT

- Fails completely on tall tables — produces empty outputs
- Loses column-span hierarchy in multi-level headers
- Dense visual layouts exceed visual encoding capacity

TableSight: Deployment Dashboard

4 Dashboard Tabs

Compare

Single-image inference — upload or pick random from 1,300 gallery images. Models run in parallel, TEDS shown side-by-side.

Benchmark

Pre-computed aggregate stats over 1,300 samples, broken down by image_type and difficulty.

Batch Eval

Run all models over uploaded CSV slice, export per-sample scores.

Failure Analysis

Surfaces all samples with TEDS < 0.5 with diff vs ground truth.

HTML → DataFrame Export

Parser Input

Model-generated HTML (any model)

colspan expand

Repeat header across spanned columns

rowspan carry

Carry merged cells into following rows

Pad short rows

Normalize to rectangular shape

Download CSV

1,495/1,510 complex samples: clean parse

Key Findings

1

Structure ≠ Content

Best structure model (UniTable, 0.8353) ≠ best content model (SPARTAN, 0.5391 Pipeline TEDS). Optimize separately.

2

Cascading Errors Dominate

Multi-stage pipelines amplify small mistakes. SPARTAN's local error confinement beats UniTable's global decode despite weaker structure.

Future Work & Recommendations

Direction 1: Fine-Tuning UniTable with more epochs and higher learning rate.

Fine-tune the content model using more epochs, higher learning rate and increase the impact of LoRA.

Direction 2: Learned Row-Grouping Classifier

Replace SPARTAN's fixed row_tol_frac threshold with a learned classifier. Directly attacks span loss (54.2% of failures) without hurting multi-line wrapped layouts.

Direction 3: Hybrid Architecture

Recommended: UniTable skeleton + SPARTAN OCR + parsing layer. Compose model strengths rather than demanding one model own the whole pipeline.